

Quiz-1

ELL7282/ELL782: Introduction to Computer Architecture

IIT Delhi · Semester I, 2026–27

(Answer all the questions with proper explanations.)

Total Marks: 50

Time: 60 minutes

Instructions: Answer all questions with necessary reasoning. Truth tables/K-maps are not allowed in Q1. For IEEE 754 use single precision with round-to-nearest, ties-to-even. In Q5, **sbn a,b,L** means $a \leftarrow a - b$ and branch to L iff the new $a < 0$. SimpleRisc operations are 32-bit; default immediates are sign-extended, suffix **u** zero-extends, and suffix **h** places the immediate in bits 31:16.

1. Consider

$$F(A, B, C, D) = AB + A'C + BC + AD + A'D + CD,$$

where $A, B, C, D \in \{0, 1\}$, $+$ denotes OR, juxtaposition denotes AND, and X' denotes NOT X .

- (a) Simplify F using Boolean algebra only. Clearly mention all laws you use. [4]
- (b) Realize the simplified function using the minimum possible number of 2-input NAND gates. State the minimum gate count, name all intermediate signals, and justify why fewer gates are insufficient. [4]

2. In an unknown radix r ,

$$(57.3)_r + (24.6)_r = (104.1)_r,$$

where $(X)_r$ denotes the numeral X written in base r , and the point is the radix point. Thus, for example,

$$(57.3)_r = 5r + 7 + \frac{3}{r}.$$

- (a) Determine r algebraically and prove that it is the unique valid integer radix. Recall that r must exceed every digit used. [3]
- (b) Let

$$N = (57.3)_r - (24.6)_r.$$

Find N in decimal. Assume a two's-complement fixed-point representation with exactly three fractional bits. Determine the minimum total word length required to represent both $+N$ and $-N$, and give the bit patterns for $+N$, $-N$, and the 16-bit sign extension of $-N$.

The binary point is assumed to lie immediately before the three least-significant bits. [3]

- (c) In an n -bit two's-complement fixed-point system with f fractional bits, let $F_f(u)$ denote the unsigned integer value of the n -bit pattern representing u . The represented numerical value is therefore the signed integer associated with the pattern divided by 2^f . Prove

$$F_f(-u) \equiv 2^n - F_f(u) \pmod{2^n}.$$

Hence explain why negating the most-negative representable fixed-point number produces the same bit pattern, although the mathematical positive value is different. [3]

3. Let $x = 2^{24}$ and $y = 1$ be IEEE 754 single-precision numbers. Here, $\text{fl}(z)$ denotes z rounded to single precision using round-to-nearest, ties-to-even.

- (a) Give the complete 32-bit hexadecimal encodings of x and y , and the numerical distance from x to the next larger representable number. [2]
- (b) Evaluate, showing each non-trivial rounding decision,

$$E_1 = \text{fl}(\text{fl}(x + y) - x), \quad E_2 = \text{fl}(x + \text{fl}(y - x)),$$

where E_1, E_2 are the final floating-point results. Explain why $E_1 \neq E_2$. [3]

- (c) Find the greatest representable number below x , its hexadecimal encoding, and explain why the representable-number gaps immediately below and above 2^{24} differ. [2]

4. A three-dimensional array

`short A[5][7][9]`

starts at byte address 0x8000; each element occupies 2 bytes. An element starts at 0x81BC. Here, $A[i][j][k]$ denotes the element indexed by i, j, k ; the rightmost index varies fastest in row-major order, and the leftmost varies fastest in column-major order.

- (a) Determine (i, j, k) under (i) row-major and (ii) column-major organization. Show the byte-address calculation. [4]
- (b) If the element stores decimal -300 , give its 16-bit two's-complement representation and the bytes at 0x81BC and 0x81BD for little endian (least-significant byte first) and big endian (most-significant byte first). [2]
- (c) If $r1 = 0x8000$ and $r2 = 0x1A0$, determine the addressing mode, effective address

$$EA = r1 + r2 + 28$$

(where 28 is decimal), and the corresponding row-major array element for $28[r1, r2]$. [1]

- (d) A 32-bit word 0x00613041 is stored at 0x81C0 on a little-endian machine. List its bytes in increasing-address order and give the NUL-terminated ASCII string, where NUL is 0x00. Are the printable-character byte values identical in UTF-8? [1]

5. The machine has only

`sbn a,b,L :` $a \leftarrow a - b$; branch to L iff the new $a < 0$,

where a, b are memory operands and L is a label. Initially,

$$x \geq 0, \quad y > 0, \quad q = 0, \quad t = 0, \quad \text{one} = 1,$$

where x_0 denotes the initial value of x , q is the quotient, t is temporary storage, and **one** stores constant 1.

- (a) Write an SBN-only program terminating with

$$q = \left\lfloor \frac{x_0}{y} \right\rfloor, \quad x = x_0 \bmod y.$$

Preserve y and assume no other initialized constants. [6]

- (b) State a loop invariant, meaning a relation true before every iteration, and give a termination argument proving the final values of q and x . [2]

- 6. (a) A processor has fixed 32-bit instructions, register-to-register ALU operations, explicit load/store memory access, and one unified byte-addressed memory. Here, ALU means arithmetic logic unit; RISC/CISC describe ISA styles; Harvard/Von Neumann describe memory organization. Classify the ISA and memory organization, and explain briefly how a large register file exploits locality. [2]

- (b) Register **r0** contains a 32-bit word x , **r9** contains a destination byte address, and **0[r9]** denotes memory at address $r9 + 0$. Write a straight-line SimpleRisc program, with no loop or function call, that reverses

$$B_3B_2B_1B_0 \longrightarrow B_0B_1B_2B_3,$$

leaves the result in **r1**, stores it at **0[r9]**, and preserves **r0, r9**. Use at least one immediate modifier: **u** means zero extension and **h** places the immediate in bits 31:16.

For $x = 0x12345678$, state the final **r1**. Also explain why logical right shift, which inserts zeros, is required instead of arithmetic right shift, which replicates the sign bit. [8]

— End of Question Paper —